

Learning Global Illumination Structure for Multi-Source Spatial Inversion

1 Research Objective

We study a spatial inverse problem in which the scene geometry is known, the underlying light-source configuration is unknown, and illumination is observed only within a local region of the scene.

Given scene geometry S , a spatially localized observation I_{obs} , and its observation mask M_R , the model aims to infer three complementary quantities:

$$(S, I_{\text{obs}}, M_R) \longrightarrow \left(\hat{P}_{\text{src}}(\mathbf{x}), \hat{I}(\mathbf{x}), \{\hat{P}_{\text{block}}(o_j)\}_{o_j \in S} \right), \quad \mathbf{x} \in \Omega. \quad (1)$$

Here,

- $\hat{P}_{\text{src}}(\mathbf{x})$ describes the probability that a light source is located near $\mathbf{x}$;
- $\hat{I}(\mathbf{x})$ is the predicted global illumination field;
- $\hat{P}_{\text{block}}(o_j)$ describes the probability that obstacle point o_j contributes to blocking illumination reaching the current observation region.

A global visibility partition and its corresponding changing surface are further derived from the predicted illumination field through a learned threshold.

2 Physical Formulation

Let the spatial domain be

$$\Omega \subset \mathbb{R}^d, \quad d \in \{2, 3\}. \quad (2)$$

A scene contains obstacle geometry S and K light sources

$$U = \{L_1, \dots, L_K\}. \quad (3)$$

Each source L_k has a spatial position and strength a_k .

For a spatial location q , source-specific geometric visibility is

$$V(S, q, L_k) \in \{0, 1\}, \quad (4)$$

where $V = 1$ indicates an unobstructed propagation path and $V = 0$ indicates geometric occlusion.

The illumination response is modeled as

$$I(q) = \sum_{k=1}^K V(S, q, L_k) a_k A(L_k, q), \quad (5)$$

where $A(L_k, q)$ denotes attenuation between the source and query location.

This formulation separates geometric occlusion from the continuous illumination response generated by multiple sources.

3 Synthetic Scene Generation

3.1 Random Obstacle Geometry

Scene structures are generated using spatially clustered random point sets rather than fixed geometric templates.

For cluster center c_j , obstacle points are sampled according to

$$p_{j,n} \sim \mathcal{D}(c_j, \Sigma_j), \quad (6)$$

where Σ_j controls the spatial extent and orientation of the cluster.

The complete obstacle structure is represented as

$$S = \bigcup_{j=1}^M \{p_{j,1}, \dots, p_{j,N_j}\}. \quad (7)$$

The resulting structures are converted into geometric representations suitable for ray-intersection and visibility simulation.

3.2 2D Prototype and 3D Extension

Initial experiments are conducted in

$$\Omega_{2D} = [0, L_x] \times [0, L_y]. \quad (8)$$

The same formulation is then extended to

$$\Omega_{3D} = [0, L_x] \times [0, L_y] \times [0, L_z]. \quad (9)$$

In three dimensions, a light source is represented as

$$L_k = (x_k, y_k, z_k, a_k). \quad (10)$$

The simulator generates the complete illumination field, source-specific visibility information, and blocking-object correspondences.

4 Position-Aware Local Observation

Only a local spatial region

$$R \subset \Omega \quad (11)$$

is observed.

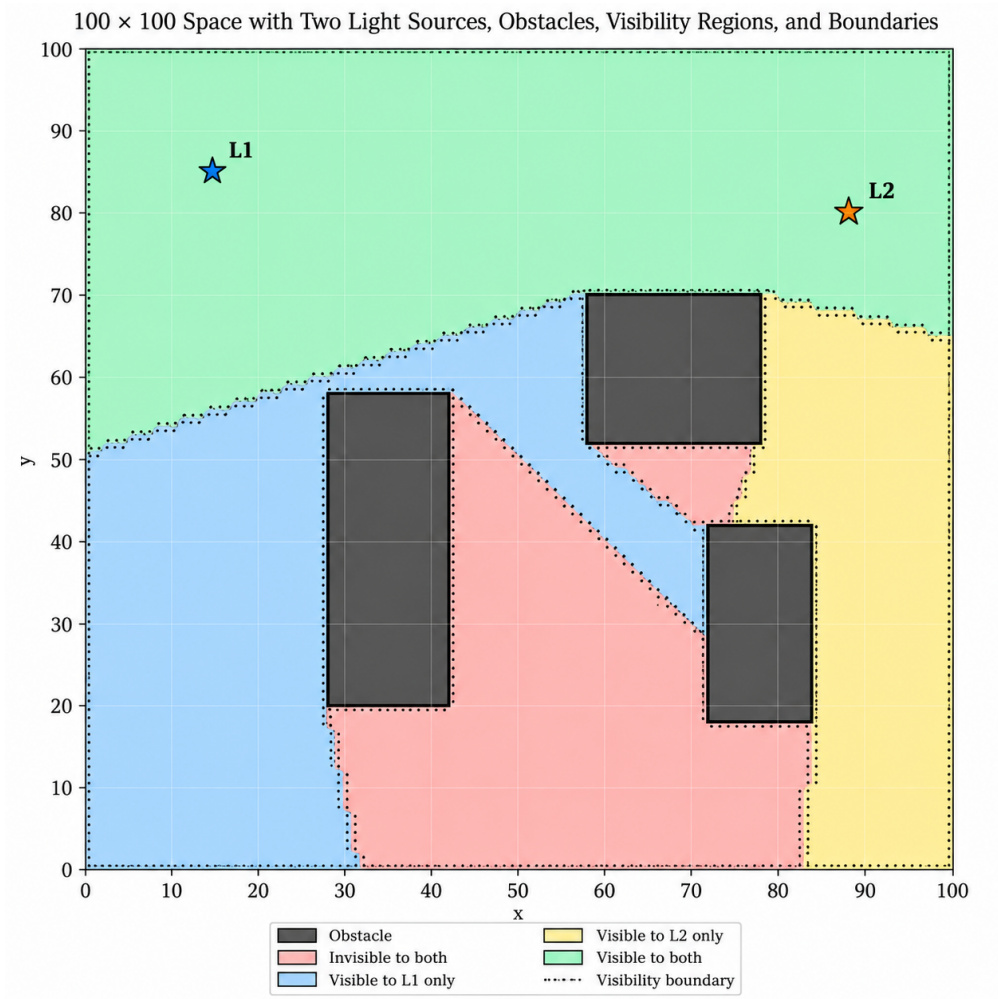


Figure 1: Example of a two-dimensional scene with multiple obstacles and two light sources. Geometric occlusion produces spatial regions with different illumination and visibility states.

The observation is embedded in the global coordinate system as

$$I_{\text{obs}}(\mathbf{x}) = \begin{cases} I(\mathbf{x}), & \mathbf{x} \in R, \\ 0, & \mathbf{x} \notin R, \end{cases} \quad (12)$$

with observation mask

$$M_R(\mathbf{x}) = \begin{cases} 1, & \mathbf{x} \in R, \\ 0, & \mathbf{x} \notin R. \end{cases} \quad (13)$$

The mask distinguishes unobserved locations from genuinely low-intensity measurements and preserves the position and extent of the observed region within the global scene.

Observation regions are randomized in position and size during data generation so that the model learns relative relationships among observations, obstacles, and light sources.

5 Prediction Targets

5.1 Source Probability Field

The source probability field is defined over the complete spatial domain:

$$P_{\text{src}}(\mathbf{x}) = P(\text{source near } \mathbf{x} \mid S, I_{\text{obs}}, M_R). \quad (14)$$

Multiple spatial modes may coexist, allowing the representation to support multi-source configurations and ambiguous inverse solutions.

5.2 Global Illumination Field

The model predicts the expected illumination throughout the complete domain:

$$\hat{I}(\mathbf{x}) \approx \mathbb{E}[I(\mathbf{x}) \mid S, I_{\text{obs}}, M_R]. \quad (15)$$

This provides a continuous global reconstruction of the physical response beyond the observed region.

5.3 Observation-Conditioned Blocking Probability

Let

$$S = \{o_1, o_2, \dots, o_N\} \quad (16)$$

denote the set of obstacle points in the scene.

For each obstacle point o_j , the model predicts

$$P_{\text{block}}(o_j) = P(o_j \text{ contributes to blocking illumination reaching } R \mid S, I_{\text{obs}}, M_R). \quad (17)$$

The resulting obstacle-wise prediction is

$$\{P_{\text{block}}(o_j)\}_{j=1}^N. \quad (18)$$

This representation identifies which parts of the known obstacle geometry are relevant to the current observed response, rather than measuring whether an obstacle contributes to occlusion somewhere in the entire scene.

6 Learned Visibility Structure

Visibility-related spatial structure is derived from the predicted global illumination field using a learned threshold.

The threshold is predicted from the scene and observation:

$$\hat{\tau} = h_{\phi}(S, I_{\text{obs}}, M_R). \quad (19)$$

A differentiable soft visibility representation is defined as

$$\hat{P}_{\text{vis}}(\mathbf{x}) = \sigma \left(\frac{\hat{I}(\mathbf{x}) - \hat{\tau}}{T} \right), \quad (20)$$

where T controls transition sharpness.

A discrete spatial partition is obtained by

$$\hat{Y}_{\text{vis}}(\mathbf{x}) = \mathbf{1} \left[\hat{I}(\mathbf{x}) > \hat{\tau} \right]. \quad (21)$$

The corresponding visibility-changing surface is

$$\hat{\Gamma}_{\text{vis}} = \partial \left\{ \mathbf{x} : \hat{Y}_{\text{vis}}(\mathbf{x}) = 1 \right\}. \quad (22)$$

Thus,

$$\hat{I} \longrightarrow \hat{\tau} \longrightarrow \hat{Y}_{\text{vis}} \longrightarrow \hat{\Gamma}_{\text{vis}}. \quad (23)$$

The simulator-generated geometric visibility information provides a physical reference for evaluating the derived visibility partition and changing surface.

7 Learning Objective

Given the scene geometry and a position-aware local observation, the model predicts

$$(S, I_{\text{obs}}, M_R) \longrightarrow \left(\hat{P}_{\text{src}}(\mathbf{x}), \hat{I}(\mathbf{x}), \{\hat{P}_{\text{block}}(o_j)\}_{o_j \in S} \right). \quad (24)$$

The source prediction estimates the spatial distribution of possible light sources, while $\hat{I}$ reconstructs the global illumination field. For each obstacle point o_j , $\hat{P}_{\text{block}}(o_j)$ estimates its contribution to blocking illumination reaching the current observation region.

In addition, the model learns a scene-conditioned illumination threshold

$$\hat{\tau} = h_{\phi}(S, I_{\text{obs}}, M_R), \quad (25)$$

which converts the predicted illumination field into a soft visibility representation

$$\hat{P}_{\text{vis}}(\mathbf{x}) = \sigma \left(\frac{\hat{I}(\mathbf{x}) - \hat{\tau}}{T} \right). \quad (26)$$

The threshold is learned jointly with the other prediction targets by comparing the derived visibility field with simulator-generated visibility supervision. Therefore, the model learns both the continuous illumination field and the decision level at which its spatial visibility structure emerges.

Conceptually, the training objective contains four components:

$$\mathcal{L} = \lambda_{\text{src}}\mathcal{L}_{\text{src}} + \lambda_{\text{illum}}\mathcal{L}_{\text{illum}} + \lambda_{\text{block}}\mathcal{L}_{\text{block}} + \lambda_{\text{vis}}\mathcal{L}_{\text{vis}}. \quad (27)$$

Here, $\mathcal{L}_{\text{vis}}$ provides supervision for the learned threshold through the derived visibility field. The exact loss functions and weights will be determined experimentally.

8 3D Representation

After discretizing the spatial domain into an $H \times W \times D$ grid,

$$\hat{P}_{\text{src}} \in [0, 1]^{H \times W \times D}, \quad (28)$$

$$\hat{I} \in \mathbb{R}_{\geq 0}^{H \times W \times D}. \quad (29)$$

Obstacle attribution is represented separately over the N obstacle points:

$$\hat{\mathbf{p}}_{\text{block}} = \left[\hat{P}_{\text{block}}(o_1), \dots, \hat{P}_{\text{block}}(o_N) \right]. \quad (30)$$

These outputs can be visualized jointly within the same three-dimensional scene.

For example, obstacle points can be colored according to their predicted blocking probabilities while the source field, illumination field, observation region, and visibility-changing surfaces are displayed simultaneously.

9 Experimental Questions

The initial study focuses on the following questions:

- (1) Can a local illumination observation recover the global spatial distribution of hidden light sources?
- (2) Can the global illumination field be reconstructed accurately outside the observed region?
- (3) Can the learned threshold recover meaningful visibility regions and changing surfaces?
- (4) Can the model identify which obstacle points contribute to blocking illumination reaching the observed region?
- (5) How important is the global position of the local observation for inverse reasoning?
- (6) Do these representations generalize to previously unseen scene geometries?

10 Evaluation

Source inference will be evaluated by comparing the predicted source probability field with ground-truth source locations.

Global illumination prediction will be evaluated by comparing

$$\hat{I}(\mathbf{x}) \quad (31)$$

with the simulator-generated illumination field $I(\mathbf{x})$, particularly outside the observation region.

The derived visibility partition and changing surface will be compared with simulator-generated visibility information.

Blocking attribution will be evaluated by comparing

$$\hat{P}_{\text{block}}(o_j) \tag{32}$$

with the observation-specific target

$$Y_{\text{block}}^R(o_j). \tag{33}$$

Generalization will be evaluated using scene-level splits with unseen obstacle structures and randomized observation regions.

Key ablations include:

- position-aware versus position-agnostic observations;
- fixed versus learned illumination threshold;
- source and illumination prediction with versus without obstacle-blocking supervision.

11 Expected Contributions

The proposed study investigates local-to-global inverse reasoning from partial illumination observations.

The expected contributions are:

- (1) probabilistic spatial inference of multi-source configurations;
- (2) global illumination reconstruction from localized observations;
- (3) learned extraction of visibility regions and changing surfaces;
- (4) observation-conditioned attribution of blocking responsibility to individual obstacle points;
- (5) a synthetic 2D/3D dataset containing source, illumination, visibility, and source–obstacle–response correspondence annotations;
- (6) evaluation of transfer across previously unseen spatial geometries.